Land Alchemist

Project Team 150

Presented by

Nick Heppermann

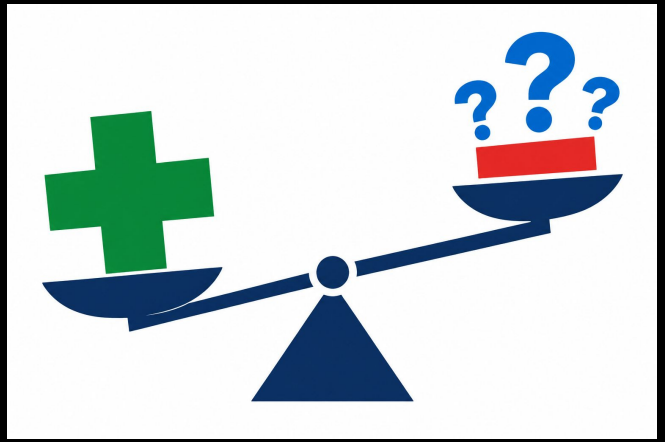
Tharun Sai Anantha

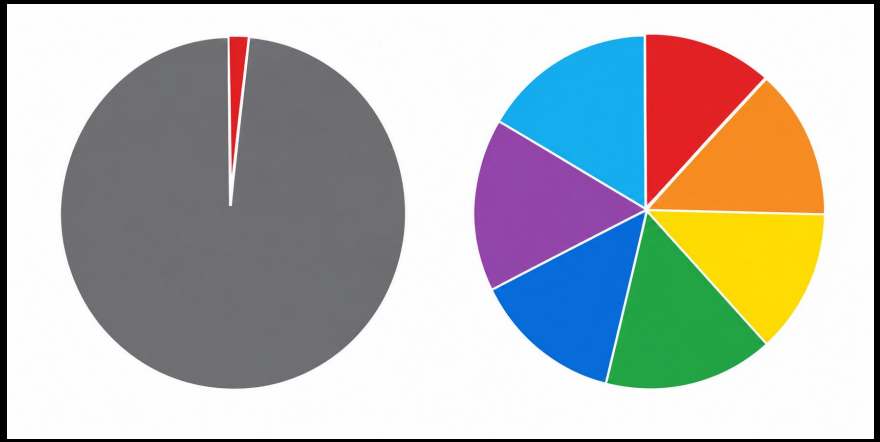
Anik Mahmood

Andrei Spatariu

What is Land Alchemist and Why it Exists?

- Data and visualization platform
- Analyze features external to homes
- Simulate scenarios





How It's Done Today & The Gap

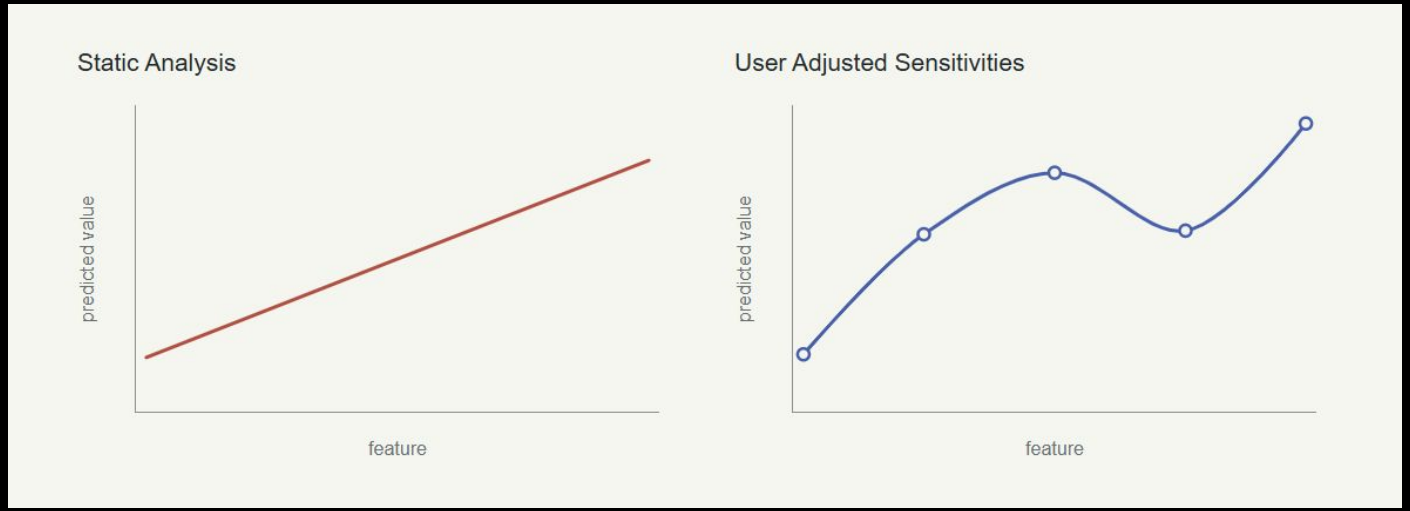
- Studies look at one feature at a time
  - Negatives lower prices: landfills, toxic plants, power plants, contamination [1, 3, 5, 6, 12]
  - Positives raise prices: schools, colleges, hospitals, rail, amenities [2, 4, 7, 11]
- Effects differ by place and by person [2, 10]
- Existing tools hide negatives and give no forecast [9], or don't let users weigh features [8]
- Gap: no tool puts good and bad features together with user-set sensitivities

References:
[1] Bouvier 2000 · [2] Chen 2022 · [3] Currie 2015 · [4] Davidoff 2008 · [5] Davis 2011 · [6] Hite 2001 · [7] Karanikolas 2026 · [8] Kmen 2026 · [9] Li (HomeSeeker) 2018 · [10] Muehlenbachs 2015 · [11] Rivas 2019 · [12] Zwickle 2019


Make the evaluation process
personal and apparent.

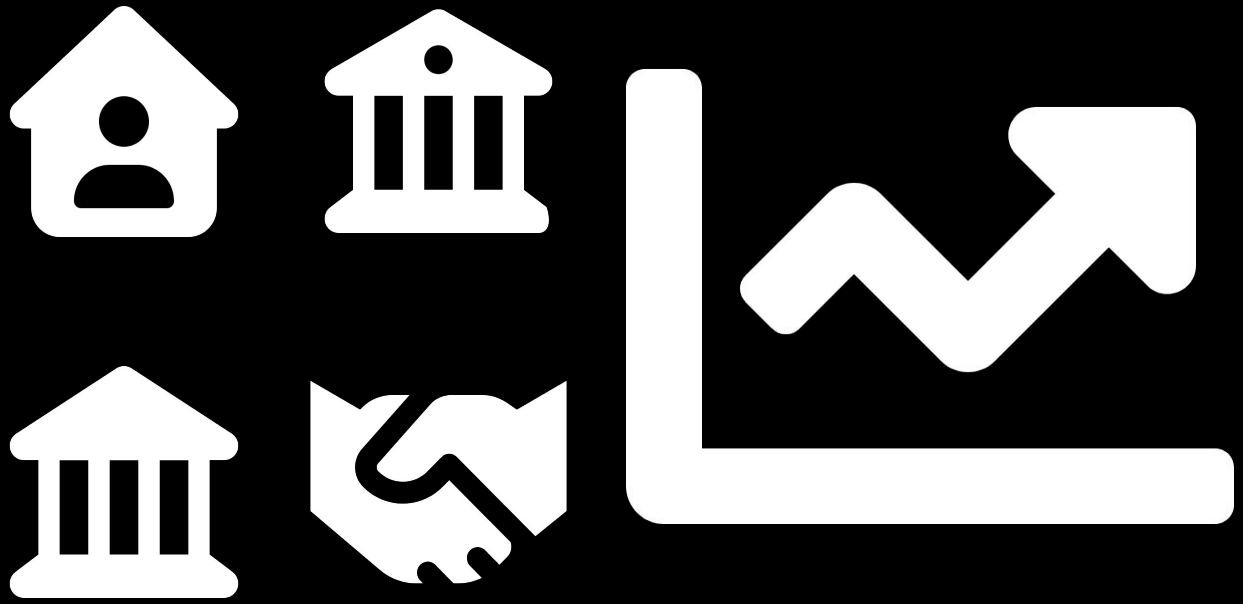
Land Alchemist Capabilities

- Baseline sensitivities determined by Land Alchemist
- Users inject their biases for personalized experience



Land Alchemist Target Audience

- Buyers and sellers
- Lenders
- Government officials



- Payoffs
  - Assists all with price discovery ($10^6 - $10^7)
- Risks
  - Opportunity costs of other endeavors

Plan of Activities & Cost

Activity	Milestone	Start Date	Expected End Date	Status	Team Members	Gantt Chart
Proposal Documentation	-	10/2	10/15	Completed	Andrei, Anik, Nick, Tharun	
Explored data sets for 10 features	Midterm	10/9	11/6	In Progress	Andrei, Tharun	
Harvested predicted value data sets	Midterm	10/15	11/6	In Progress	Anik, Tharun	
Create mock ups of visualizations, discuss user journey	Midterm	10/25	11/6	Not Started	Andrei, Nick	
Identified multiple features to predicted value sensitivities	Midterm	10/25	11/6	Not Started	Anik, Tharun	
Create basic structure of app (backend/frontend)	Midterm	10/28	11/4	Not Started	Andrei, Nick	
Document build instructions	-	11/1	11/4	Not Started	Andrei, Anik, Nick, Tharun	
Data lake export POC and documentation	-	11/5	11/7	Not Started	Nick	
Data cleaning and joining	-	11/5	11/6	Not Started	Andrei, Anik, Nick, Tharun	
Analysis to determine static weights	-	11/5	11/6	Not Started	Nick	
Determine formula for dynamic weights	-	11/5	11/9	Not Started	Andrei, Anik, Nick, Tharun	
Application Testing	Final	11/9	12/1	Not Started	Andrei, Anik, Nick, Tharun	
Application supports user workflows, user interactivity, and produces visualizations	Final	11/1	12/4	Not Started	Andrei, Anik, Nick, Tharun	
User testing and feedback	Final	12/1	12/4	Not Started	Nick	
Poster Preparation	Final	12/1	12/4	Not Started	Andrei, Anik, Nick, Tharun	

Total Cost: Free, public data sets and student digital resources

The End

See you in December 2026